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Tristan Jones

Mission: Solving Complex Problems to Advance Human Well-Being

Summary: Biomedical Informatics M.S. graduate with experience in medical imaging, biomedical data analysis, machine learning, neurodiagnostics, and applied clinical research. Skilled in Python, R, imaging workflows, data preprocessing, model evaluation, and clinical research operations, with applied projects in CT atlas construction, algorithm design, computer vision, healthcare analytics, and genomic data equity.

Education

Graduated May 2026 M.S. **Biomedical Informatics**, University of Nebraska at Omaha – Omaha, NE | GPA: 3.64

- Graduate training in biomedical data science, machine learning, medical imaging, algorithm design, healthcare analytics, clinical decision support, and translational research workflows.
- Capstone: Created a probabilistic CT liver atlas to estimate distances to the major hepatic and portal vasculature. Required medical image registration, segmentation outputs, anatomical variability analysis, and clinically motivated computational pipeline design; improved median Dice similarity from 0.146 before alignment to 0.724 after alignment across 261 subjects.

Graduated June 2022 B.S. **Bioengineering: Biosystems**, UC San Diego – La Jolla, CA | GPA: 3.2

- Interdisciplinary training in biological systems, engineering analysis, physiology, modeling, signal processing, instrumentation, and biomedical technology design.
- Capstone: Laryngospasm Device - collaborated on an interdisciplinary senior design team mentored by a throat surgeon to develop and test a device concept for inhibiting laryngospasm in the operating room; contributed raw data analysis, signal processing, calibration, parameterization, and metrology-oriented testing.

Relevant Work Experience

Sept 2025 – May 2026 **Graduate Student Researcher**, University of Nebraska Omaha – Omaha, NE

- Developed methods for spatial transformation and registration of heterogeneous CT scans, enabling alignment of individual patient images into a shared reference space.
- Built and maintained Python-based preprocessing and analysis pipelines for medical imaging data, including normalization, spatial harmonization, and quality assessment.
- Conducted computational analysis of clinical CT imaging data to support imaging-derived biomarker development and downstream translational research applications.
- Designed standardized, reproducible preprocessing workflows to improve data consistency, quality control, and analytical reliability across imaging datasets.

Nov 2022 – Dec 2024 **MEG Technologist and Research Coordinator**, Nebraska Medicine/UNMC – Omaha, NE

- Supported database-driven biomedical research for studies involving Alzheimer's disease, epilepsy, and sleep disorders through structured data cleaning, transcription, and ETL-style workflows for multimodal datasets, including actigraphy-based sleep metrics.
- Performed high-quality clinical and research MEG and 64-channel EEG acquisitions for epilepsy and brain tumor patients in a hospital setting, ensuring data integrity and protocol compliance.
- Conducted participant-facing research activities, including informational interviews, informed consent discussions, and administration of cognitive assessments in accordance with IRB-approved protocols.
- Managed IRB protocol submissions, amendments, and regulatory documentation through institutional Research Support Systems (RSS).

Technical Skills

Programming & Algorithms: Python, R, MATLAB, NumPy, pandas, SciPy, dplyr, ggplot2, data structures & algorithms, time/space complexity
ML/DL: scikit-learn, TensorFlow/Keras, random forest, XGBoost, KNN, PCA/eigenfaces, CNNs, autoencoders, VAEs, GANs, Grad-CAM, cross-validation, ROC/AUROC
Imaging & Signal Data: CT, MRI, MEG, EEG, NIfTI, DICOM, TotalSegmentator, registration, segmentation, spatial normalization
Clinical/Research: ETL workflows, data cleaning, QC, IRB documentation, clinical data governance, human subjects research
Project Management: Trello, Kanban boards, sprint planning, task tracking, workflow documentation, reproducible pipeline design

Selected Project Experience

Sept 2025 – May 2026 **Probabilistic CT Liver Atlas for Distance-to-Vasculature Estimation**, Graduate Capstone

- Developed an end-to-end medical imaging pipeline using CT volumes, TotalSegmentator outputs, NIfTI affine normalization, rigid centroid-based alignment, probabilistic atlas construction, and distance-to-vasculature mapping.
- Created sex-stratified liver atlases across 261 subjects, improving median Dice similarity from 0.146 before alignment to 0.724 after alignment and generating population-level hepatic/portal vasculature distance maps.

March – May 2026 **Nearest-Neighbor Search Algorithms: Brute Force, KD-Tree, and ANN**

- Implemented brute-force search, exact k-d tree search, and randomized k-d tree forest approximate nearest-neighbor methods from scratch in Python.
- Benchmarked runtime, memory usage, and search accuracy across synthetic clustered datasets of 100,000 points and dimensionalities from 2 to 200, emphasizing algorithmic tradeoffs and the curse of dimensionality.

Jan – May 2026 **Interpretable Machine Learning for K–12 Student Outcome Prediction**

- Contributed to a team-based predictive modeling project using real-world institutional high school data, supporting workflow design, data interpretation, model evaluation, and documentation.
- Interpreted results from random forest, XGBoost, SMOTE-based class balancing, and pruned decision tree models using ROC AUC, sensitivity, specificity, feature importance, partial dependence plots, and individual-level waterfall explanations.

Nov – Dec 2025 **Variational Autoencoder for Image Anomaly Detection**

- Implemented a convolutional variational autoencoder on Fashion-MNIST using a probabilistic latent space, reparameterization trick, reconstruction loss, and KL-divergence regularization.
- Evaluated anomaly detection using reconstruction error, AUROC, confusion matrices, 95th-percentile thresholding, and controlled corruption testing across noise, occlusion, rotation, and combined anomaly types.

Oct – Nov 2025 **Genomic Data Equity, Polygenic Risk Scores, and Ancestry Representation**

- Researched how Eurocentric GWAS training data, linkage disequilibrium differences, haplotype structure, and ancestry underrepresentation affect polygenic risk score transferability.
- Emphasized genomic data quality, ancestry-stratified validation, fairness-aware model evaluation, public health implications, and responsible biomedical prediction deployment.

Oct 2025 **VGG16 Image Classification and Grad-CAM Explainability**

- Used a pretrained VGG16 model with ImageNet weights to classify images and generate Grad-CAM heatmaps with TensorFlow GradientTape for visual model interpretation.
- Quantified heatmap concentration using Shannon entropy, top-10% activation focus, Gini coefficient, and activation mean/std metrics to assess model attention behavior.

Sept 2025 **Convolutional Autoencoder for CIFAR-10 Image Reconstruction**

- Built a convolutional autoencoder in TensorFlow/Keras using Conv2D, MaxPooling2D, UpSampling2D, Adam optimization, and MAE/MSE loss functions to reconstruct CIFAR-10 images.
- Compared model capacity and reconstruction quality using filter-count experiments, SSIM, PSNR, and visual reconstruction analysis, finding that increased filter capacity improved image clarity more than changing loss functions.

April – May 2025 **Facial Emotion Recognition with Eigenfaces and Machine Learning Classifiers**

- Built a facial emotion recognition pipeline using the FER2013 dataset, grayscale image preprocessing, PCA/eigenfaces dimensionality reduction, and scikit-learn classifiers.
- Compared KNN, shallow neural network, and random forest models using accuracy, runtime, cross-validation stability, inference speed, and visualization-based model evaluation.

April – May 2025 **ICU Length-of-Stay Analytics with MIMIC-IV Dataset**

- Analyzed ICU length of stay using R by joining ICU stay and admission tables, calculating timestamp-based length-of-stay values, filtering outliers, and preparing clean analytic datasets.
- Created density plots, admission-type box plots, and time-based trend visualizations with dplyr and ggplot2 to communicate healthcare resource-planning insights.